

Problem

Obtain the power factor and the apparent power of a load whose impedance is $\mathbf{Z} = 60 + j40 \, \Omega$ when the applied voltage is $v(t) = 155.56 \cos(377t + 10^\circ) \, \text{V}$.

Step-by-step solution

Step 1 of 4

Consider the value of the applied voltage $v(t)$.

$$v(t) = 155.56 \cos(377t + 10^\circ) \, \text{V}$$

Convert to complex form.

$$\mathbf{V} = 155.56 \angle 10^\circ \, \text{V}$$

Therefore, the value of the voltage in complex form is, $155.56 \angle 10^\circ \, \text{V}$.

Step 2 of 4

Calculate the value of the current $\mathbf{I}$.

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}}$$

Substitute $155.56 \angle 10^\circ \, \text{V}$ for $\mathbf{V}$ and $60 + j40 \, \Omega$ for $\mathbf{Z}$.

$$\begin{aligned} \mathbf{I} &= \frac{155.56 \angle 10^\circ}{60 + j40} \\ &= \frac{155.56 \angle 10^\circ}{72.11 \angle 33.69^\circ} \\ &= 2.157 \angle -23.69^\circ \, \text{A} \end{aligned}$$

Therefore, the value of the current is $\mathbf{I}$ is, $2.157 \angle -23.69^\circ \, \text{A}$.

Step 3 of 4

Calculate the value of the power factor pf at the load.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 10° for θ_v and -23.69° for θ_i .

$$\begin{aligned} \text{pf} &= \cos(10^\circ - (-23.69^\circ)) \\ &= \cos(10^\circ + 23.69^\circ) \\ &= \cos(33.69^\circ) \\ &= 0.8321 \text{ lagging} \end{aligned}$$

Lagging since θ_i lags behind θ_v .

Therefore, the value of the power factor pf at the load is, 0.8321 lagging .

Step 4 of 4

Calculate the value of the apparent power S absorbed by the load.

$$S = \left(\frac{V_m}{\sqrt{2}} \right) \left(\frac{I_m}{\sqrt{2}} \right)$$

Substitute 155.56 V for V_m and 2.157 A for I_m .

$$\begin{aligned} S &= \left(\frac{155.56}{\sqrt{2}} \right) \left(\frac{2.157}{\sqrt{2}} \right) \\ &= \frac{335.54}{2} \\ &= 167.77 \text{ VA} \end{aligned}$$

Therefore, the value of the apparent power S is, 167.77 VA.